



American International University- Bangladesh (AIUB)

Faculty of Engineering (EEE)

Course Name :	Microprocessor and Embedded Systems	Course Code :	COE3104
Semester :	Summer 2024-25	Sec :	K
Lab Instructor :	MD. ALI NOOR		

Experiment No :	02
Experiment Name :	Familiarization with an STM32, the study of blink test and implementation of a light-controlling system using microcontrollers.

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Performance Date:	16-7-25	Due Date:	08-08-25
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Marking Rubrics (to be filled by Lab Instructor)

Category	Proficient [6]	Good [4]	Acceptable [2]	Unacceptable [1]	Secured Marks
Theoretical Background, Methods & procedures sections	All information, measures and variables are provided and explained.	All Information provided that is sufficient, but more explanation is needed.	Most information correct, but some information may be missing or inaccurate.	Much information missing and/or inaccurate.	
Results	All of the criteria are met; results are described clearly and accurately;	Most criteria are met, but there may be some lack of clarity and/or incorrect information.	Experimental results don't match exactly with the theoretical values and/or analysis is unclear.	Experimental results are missing or incorrect;	
Discussion	Demonstrates thorough and sophisticated understanding. Conclusions drawn are appropriate for analyses;	Hypotheses are clearly stated, but some concluding statements not supported by data or data not well integrated.	Some hypotheses missing or misstated; conclusions not supported by data.	Conclusions don't match hypotheses, not supported by data; no integration of data from different sources.	
General formatting	Title page, placement of figures and figure captions, and other formatting issues all correct.	Minor errors in formatting.	Major errors and/or missing information.	Not proper style in text.	
Writing & organization	Writing is strong and easy to understand; ideas are fully elaborated and connected; effective transitions between sentences; no typographic, spelling, or grammatical errors.	Writing is clear and easy to understand; ideas are connected; effective transitions between sentences; minor typographic, spelling, or grammatical errors.	Most of the required criteria are met, but some lack of clarity, typographic, spelling, or grammatical errors are present.	Very unclear, many errors.	
Comments:				Total Marks (Out of):	

Familiarization with an STM32, the study of blink test and implementation of a light-controlling system using microcontrollers.

The STM32 Nucleo-F401RE Board is a versatile development board featuring the STM32F401RE microcontroller with a 32-bit ARM Cortex-M4 core. It offers flexible connectivity with Arduino Uno V3 compatibility, making it ideal for a range of applications from prototyping to advanced projects.

1. Familiarize with the STM32 microcontroller
2. Implement a simple circuit to make an LED light to blink using the delay function
3. Implement a simple traffic control system

STM32CubeIDE is an advanced C/C++ development platform with peripheral configuration, code generation, code compilation, and debug features for STM32 microcontrollers and microprocessors. STM32CubeIDE includes build and stack analyzers that provide the user with useful information about project status and memory requirements. STM32CubeIDE also includes standard and advanced debugging features including views of CPU core registers, memories, and peripheral registers, as well as a live variable watch, Serial Wire Viewer interface, or fault analyzer.

Figure-1: STM32 Nucleo-F401RE Board [1]

NUCLEO-F401RE

Pinout diagram for the NUCLEO-F401RE board. The diagram shows the top and bottom headers with their respective pin numbers and functions. The top header (CN7, CN6, CN9, CN10) includes pins for power (VDD, GND), reset (RESET), and various I/O pins (PC10-PC13, BOOT0, NC, PA13-PA15, GND, PB7, PC13, PC14, PC15, PH0, PH1, VBAT, PC2, PC3). The bottom header (CN8, CN9, CN7) includes pins for power (VDD, GND), reset (RESET), and various I/O pins (PC10-PC13, BOOT0, NC, PA13-PA15, GND, PB7, PC13, PC14, PC15, PH0, PH1, VBAT, PC2, PC3). The diagram also shows the internal connections between the headers and the microcontroller pins.

Header	Pin	Function	Header	Pin	Function
Top	1	VDD	1	1	PC8
Top	2	GND	2	2	PC9
Top	3	VDD	3	3	PC10
Top	4	GND	4	4	PC11
Top	5	BOOT0	5	5	PC12
Top	6	NC	6	6	PC13
Top	7	NC	7	7	U5V
Top	8	NC	8	8	NC
Top	9	PA13	9	9	PA12
Top	10	PA14	10	10	PA11
Top	11	PA15	11	11	PB12
Top	12	GND	12	12	GND
Top	13	PB7	13	13	PB2
Top	14	PC13	14	14	PB1
Top	15	PC14	15	15	PB15
Top	16	PC15	16	16	PB14
Top	17	PH0	17	17	PB13
Top	18	PH1	18	18	AGND
Top	19	VBAT	19	19	PC4
Top	20	PC2	20	20	PC3
Top	21	PC3	21	21	NC
Bottom	1	VDD	1	1	PC8
Bottom	2	GND	2	2	PC9
Bottom	3	VDD	3	3	PC10
Bottom	4	GND	4	4	PC11
Bottom	5	BOOT0	5	5	PC12
Bottom	6	NC	6	6	PC13
Bottom	7	NC	7	7	U5V
Bottom	8	NC	8	8	NC
Bottom	9	PA13	9	9	PA12
Bottom	10	PA14	10	10	PA11
Bottom	11	PA15	11	11	PB12
Bottom	12	GND	12	12	GND
Bottom	13	PB7	13	13	PB2
Bottom	14	PC13	14	14	PB1
Bottom	15	PC14	15	15	PB15
Bottom	16	PC15	16	16	PB14
Bottom	17	PH0	17	17	PB13
Bottom	18	PH1	18	18	AGND
Bottom	19	VBAT	19	19	PC4
Bottom	20	PC2	20	20	PC3
Bottom	21	PC3	21	21	NC

Legend: ■ Arduino ■ Morpho

Pin configuration from STM32Cube IDE:



- 1) STM32 Cube IDE (1.0.1 or any recent version)
- 2) STM32 Cube IDE board
- 3) Breadboard
- 4) LED lights (RED, YELLOW, GREEN)
- 5) Three 220 ohms resistors
- 6) Jumper wires

A. Initial Setup

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Figure-4: Initial hardware setup of STM32 Nucleo-F401RE Board

B. Using STM32Cube IDE to write code for a simple blink program [3]

1. On a breadboard, the positive terminal of an LED was connected to the PA5/D13 pin of the STM32 Nucleo-F401RE board, and the negative terminal of the LED was connected to the GND pin alongside a 220-ohm resistor.
2. The STM32Cube IDE was opened, and from the pin configuration window, the pin PA5 was chosen as GPIO_Output.
3. The project explorer was navigated to Core > Src > main.c.
4. The program was written in the first while loop as follows.

```
96  /* Infinite loop */
97  /* USER CODE BEGIN WHILE */
98  while (1)
99  {
100     /* USER CODE END WHILE */
101     HAL_GPIO_TogglePin(GPIOA, GPIO_PIN_5); // Toggles between ON & OFF
102     HAL_Delay(1000);                        // 1 second delay
103     /* USER CODE BEGIN 3 */
104 }
105 /* USER CODE END 3 */
106 }
107
```

Figure-5: User code for blinking LED

Here, the above code performs the following operations-

- Turn LED ON for 1 second
- Turn LED OFF for 1 second
- Repeat the operations in an infinite loop

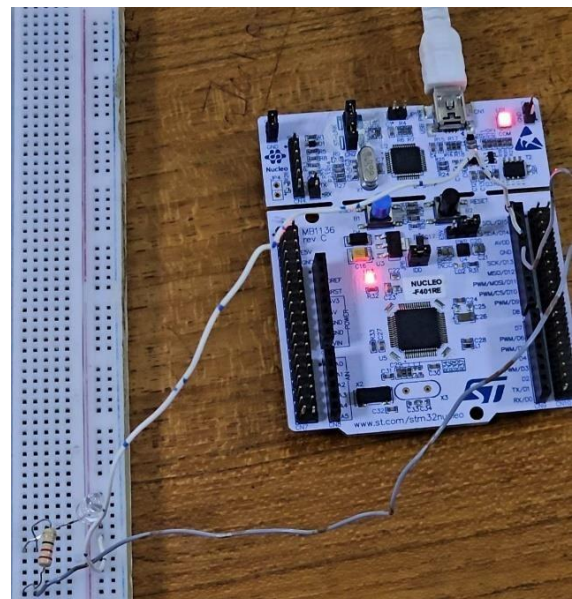
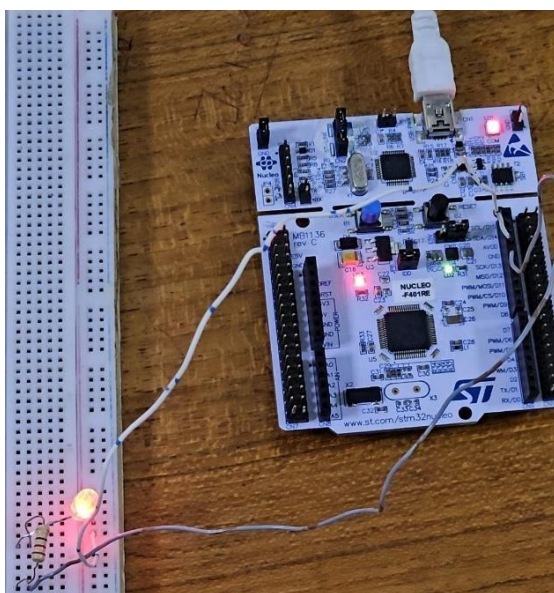


Figure-6: Hardware setup of STM32 Nucleo-F401RE Board for blink test

C. Using STM32Cube IDE to write code for a traffic light control program [4]

1. On the breadboard, the positive terminals of three LEDs (Red, Yellow, and Green) were set up, each with a 220-ohm resistor at the negative terminal. Connections were then made with the STM32 Nucleo-F401RE Board as follows:
 - The positive terminal of Red LED was connected to the PA5/D13 pin.
 - The positive terminal of Yellow LED was connected to the PA6/D12 pin.
 - The positive terminal of Green LED was connected to the PA7/D11 pin.
 - All resistors at the negative LED terminals were connected to the GND pin.
2. In the pin configuration window of the STM32Cube IDE, the pins PA5, PA6 and PA7 were all set as GPIO_Output.
3. The project explorer was again navigated to Core > Src > main.c.
4. For Traffic Light Control, the following program was written in the first while loop.

```
96  /* Infinite loop */
97  /* USER CODE BEGIN WHILE */
98  while (1)
99  {
100     /* USER CODE END WHILE */
101     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 1);           //RED LED ON
102     HAL_Delay(5000);                                     //ON for 5 second
103     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_5, 0);           //RED LED OFF
104
105     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 1);           //YELLOW LED ON
106     HAL_Delay(3000);                                     //ON for 3 second
107     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);           //YELLOW LED OFF
108
109     for(int i=0; i<3; i++) //Loop for 3 second blinking of YELLOW LED
110     {
111         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 1);       //YELLOW LED ON
112         HAL_Delay(500);                                  //ON for 0.5 second
113         HAL_GPIO_WritePin(GPIOA, GPIO_PIN_6, 0);       //YELLOW LED OFF
114         HAL_Delay(500);                                  //OFF for 5 second
115     }
116
117     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, 1);           //GREEN LED ON
118     HAL_Delay(5000);                                     //ON for 5 second
119     HAL_GPIO_WritePin(GPIOA, GPIO_PIN_7, 0);           //GREEN LED OFF
120     /* USER CODE BEGIN 3 */
121 }
122 /* USER CODE END 3 */
123 }
124
```

Figure-7: User code for Traffic Light Control Program

Here, the above code performs the following operations-

- Red LED ON for 5 seconds
- Yellow LED ON for 3 seconds
- Yellow LED Blink for 3 seconds, alternating between 0.5 seconds ON and 0.5 seconds OFF
- Green LED ON for 5 seconds

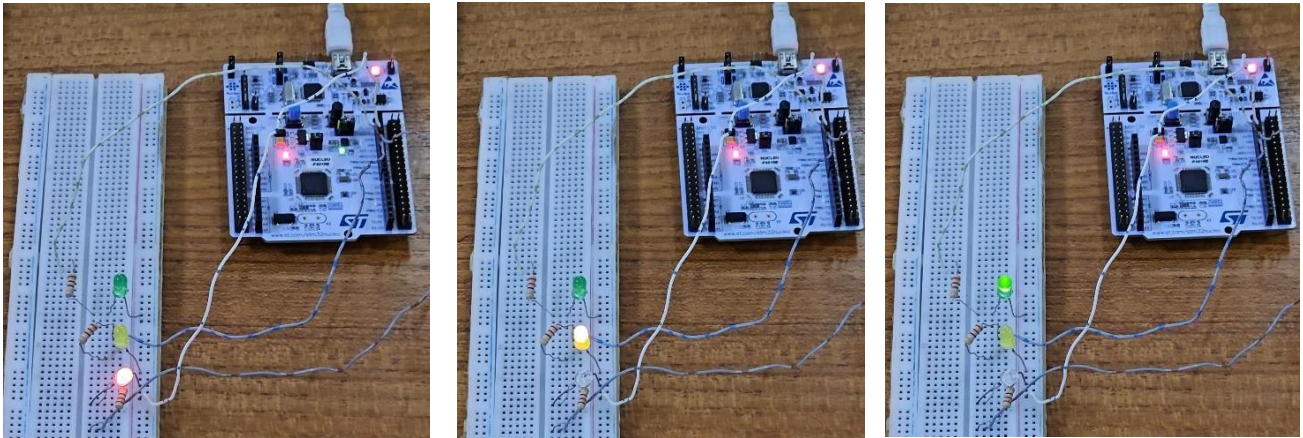


Figure-8: Hardware setup of STM32 Nucleo-F401RE Board for traffic light control system.

D. Build and debug user code from STM32Cube IDE to STM32 Nucleo-F401RE board

1. At first, the project file was saved and then built.

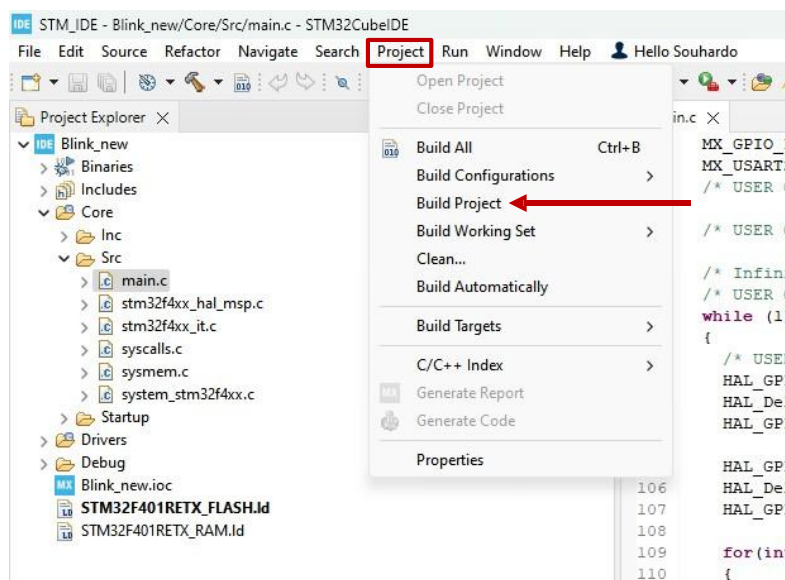


Figure-9: Build project in STM32Cube IDE

2. Next, the program was run to be debugged as an STM32 MCU C/C++ Application.

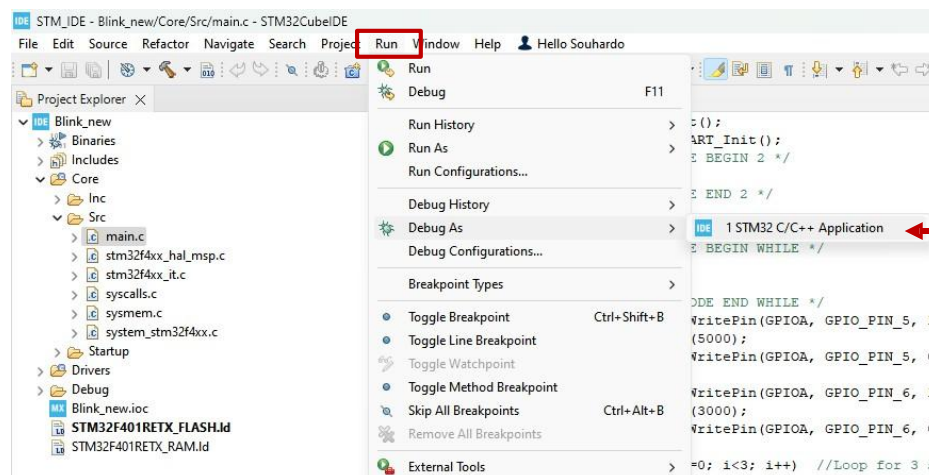


Figure-10: Debug project in STM32Cube IDE

- The resume option was selected to see the board showing results according to the user code.

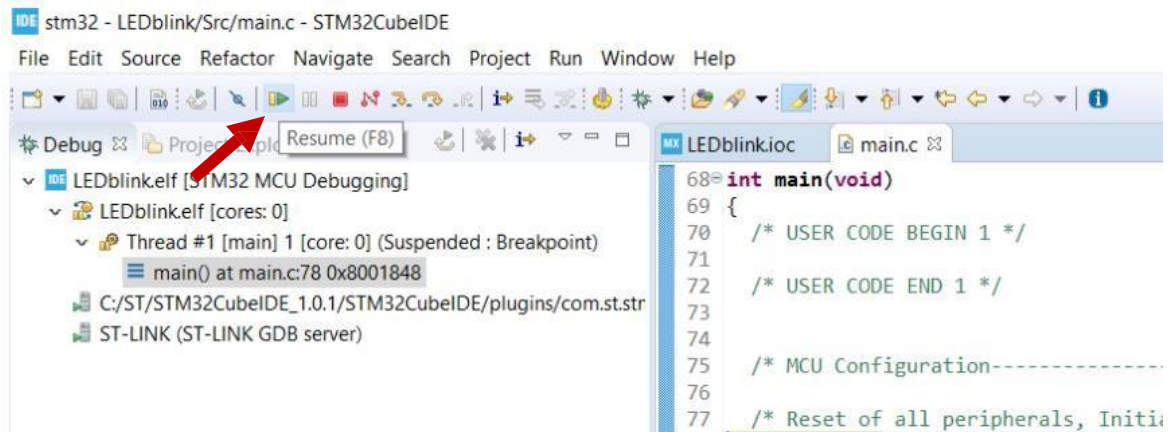


Figure-11: Run the user code to the connected STM32 Nucleo-F401RE board

Simulation:

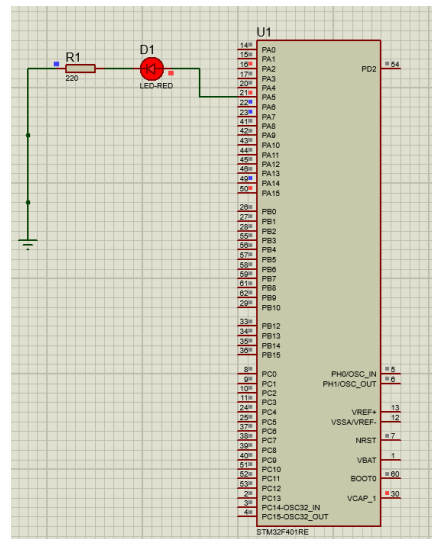


Figure-12: Simulation of blinking LED using STM32 Nucleo-F401RE

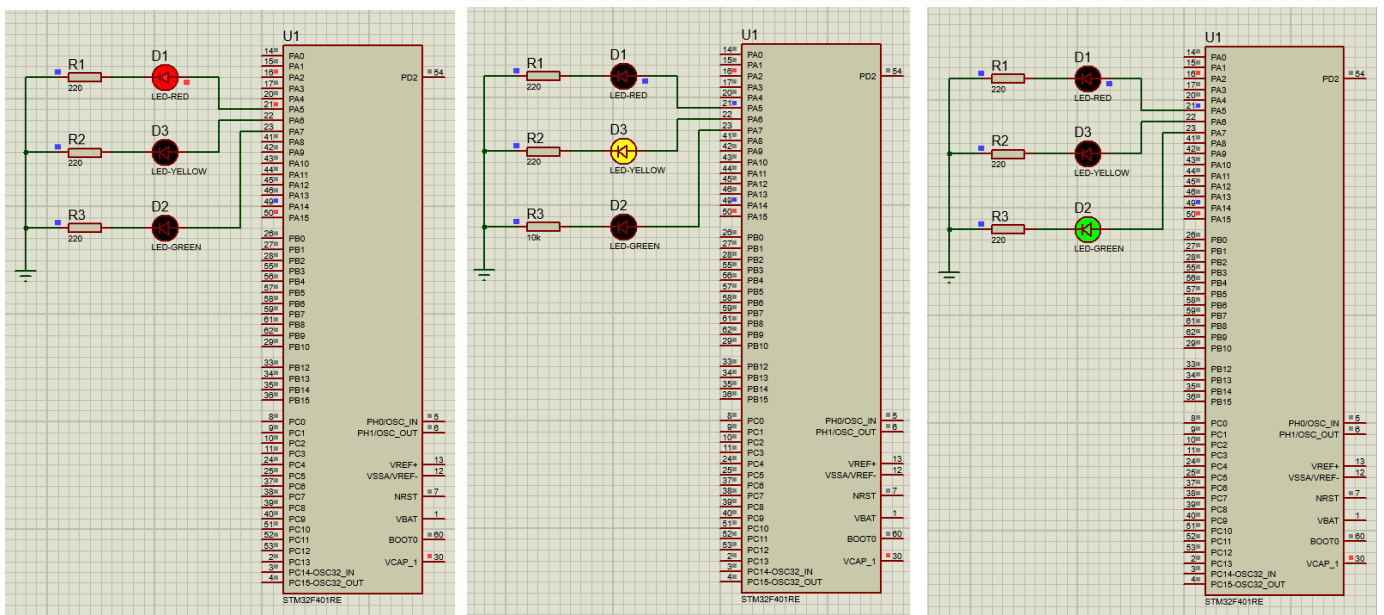


Figure-13: Simulation of traffic light control system using STM32 Nucleo-F401RE

Conclusion & Discussion:

The experiment successfully provided hands-on experience with the STM32 Nucleo-F401RE microcontroller and the STM32CubeIDE development platform, enhancing understanding of microcontroller setup and programming. In this experiment,

- The STM32 Nucleo-F401RE microcontroller was introduced, and STM32CubeIDE was used for coding and debugging.
- The blink test using a single LED demonstrated the fundamental use of GPIO pins for output and the implementation of basic timing functions
- The traffic light control program showcased the capability to manage multiple GPIO outputs simultaneously, showcasing real-world applications of microcontroller-based control systems.
- Circuits were assembled on a breadboard, emphasizing the importance of correct connections and component placement for successful testing.
- Simulations were conducted using Proteus software to verify the hardware tests. The results of the simulations showed the expected outcomes.

The experiment can be deemed as successful since no issues were encountered in either the hardware or software simulations.

Reference(s):

- [1] "STM32 Nucleo F401RE development board," *Components101*. [Online]. Available: <https://components101.com/microcontrollers/stm32-nucleo-f401re-pinout-datasheet>. [Accessed: 09-Jun-2024].
- [2] "This is information on a product in full production. STM32F401xD STM32F401xE," 2015. Available: <https://www.st.com/resource/en/datasheet/stm32f401re.pdf> [Accessed: 05-June-2024].
- [3] "STM32StepByStep:Step2 Blink LED - stm32mcu," *wiki.st.com*. https://wiki.st.com/stm32mcu/wiki/STM32StepByStep:Step2_Blink_LED [Accessed: 05-June-2024].
- [4] X. Chen and F. Chen, "Research and Design of Intelligent Traffic Signal Light Handheld Control Terminal Based on STM32," *2019 IEEE International Conference on Mechatronics and Automation (ICMA)*, Tianjin, China, 2019, pp. 1865-1870, doi: 10.1109/ICMA.2019.8816212.